PROGRESS REPORT ON THE RH AND THE EIGHT CG FUNCTIONS

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ABSTRACT. This report summarizes several aspects of our investigation into the Riemann hypothesis and the eight analogous functions of Conrey and Ghosh.

1. INTRODUCTION

For some time we have been pursuing an approach to the Riemann hypothesis based on Polya's suggestion that the function $\Phi(t)$ should play a fundamental role - see reports 2.1.5-12 [1]. Rather than repeating material, we refer the reader to the relevant portions in the reports.

The function $\Phi(t)$ is given by [JN 2.1.5 (1.3)]

$$\Phi(t) = \sum_{m=1}^{\infty} \left[2m^4\pi^2 e^{9t} - 3m^2\pi e^{5t}\right] \exp\left(-m^2\pi e^{4t}\right). \tag{1.1}$$

At the suggestion of J. B. Conrey, we have also started to use the techniques applied to the RH on a set of eight functions conjectured by Conrey and Ghosh [1] to have properties analogous to the Riemann function. The 8 CG cases are labeled by $k = 1, 2, 3, 4, 6, 8, 12, 24$, and are obtained from the Riemann case by replacing $\Phi(t)$ with $F_k(t)$ given by

$$F_k(t) = \left[\exp\left[\frac{t}{4} - \frac{\pi e^t}{12}\right] \prod_{m=1}^{\infty} (1 - e^{-2\pi m e^t})\right]^k. \tag{1.2}$$

In at least two important respects we have found that indeed there appear to be similarities between the two problems, and it is possible that a comparison of the various cases will increase understanding of all cases.

The comparison is facilitated by the use of the expansion

$$\prod_{m=1}^{\infty} (1 - y^m)^k = \sum_{j=0}^{\infty} P(j,k) y^j \tag{1.3}$$

where $y = e^{-2\pi e^t}$. For $k = 1$ the quantities $P(j,k)$ are the pentagonal numbers of Euler.

Except where noted, the remarks below concerning the RH may be applied to the 8 CG functions, if $F_k(t)$ in the form (1.3) is used to replace $\Phi(t)$ in the RH discussion.

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[1] Unpublished reports by the author are available at htpp://publish.uwo.ca/ jnuttall

2. SIGN-REGULARITY

2.1. **Rigorous results and calculations.** The heart of the method involves showing that

$$D(n,r) > 0, \quad n = 0, 1, \ldots; \ r = 1, 2..., \tag{2.1}$$

where the determinants $D(n,r)$ are given by [2.1.6 (2.7)].

- Based on $\Phi(u)$ there is a family of cumulants $\Psi_m(u)$, $m = 1, 2, \ldots$ [2.1.10 (2.8)], with corresponding kernels $K(u, v; m)$.
- If $K(u, v; m)$ kernel is sign-regular of type RR_r (see [3, p. 12]), Karlin [3] has explained how (2.1) may be verified for suitably large n, given r.
- For $r = 2$ and all n, Csordas, Norfolk and Varga [2] proved (2.1), using the concepts of compound kernels and the basic composition formula [JN 2.1.6 (Secs. 2.3 -2.6)].
- For $r = 3$, $n \geq 2$, we proved (2.1) using the same method.
- For a suitably chosen function $m(r)$ our numerical calculations suggest that $K(u, v : m(r))$ is RR_r for $r \leq 142$ [JN 2.1.9 Table 2]. Similar calculations for CG k=1 suggest an analogous result.

2.2. **Conjecture and approach to its proof.**

- We have conjectured [JN 2.1.6 p.11] that, for all $r > 1$, there exists a non-decreasing function $\mu(r)$ such that $K(u, v : \mu(r))$ is RR_r. If so, there exists a minimal function $m(r)$ such that $K(u, v; m(r))$ is RR_r.
- Results in [JN 2.1.9], [JN 2.1.11] suggest that, with large enough $\mu(r)$, only the first term of $\Phi(u)$ will be important in evaluating the kernel.
- This could lead to a proof of the conjecture.
- Given that proof, to be useful for our purposes, it will no doubt be necessary to extend the result to functions $m(r) < \mu(r)$. The ideas in [JN 2.1.12] could be helpful.

3. SINGLE MOMENTS

3.1. **Extending the proof to lower n.** Even if all the above goals are achieved, there will, for given r, be some lower values of n for which the inequality in (2.1) will remain unproved. The report [JN 2.1.7] explains how some of these lower values may be treated, assuming that the kernel $K(u, v; m)$ has the requisite sign-regularity.

- Corresponding to the kernel $K(u, v; m)$ is the two-way Wronskian $w(p, m : u)$ [JN 2.1.7 (2.12)] which satisfies [JN 2.1.7 (2.11)] when sign-regularity applies.
- In particular, in that case, we have $\Delta(n, r) = \epsilon_r w(r, m : 0) > 0$ [JN 2.1.7 (3.4)], where the determinant $\Delta(n, r)$ is given by [JN 2.1.7 (2.5)].
- Consider for example the matrix [JN 2.1.7 (3.11)] where n is even. Using the positivity of determinants $\Delta(n, r)$ for several values of (n, r), Karlin shows that the 3x3 minor of the matrix with only odd elements, i.e. $D(n, 3)$, is positive. This method allows us to fill in some of the gaps in the procedure of Sec. 2.1 above.

3.2. **Tree diagram properties.** Even when not all the required input determinants $\Delta(n, r)$ are positive, the algebraic relations underlying the method of Sec. 3.1 still apply. The report [JN 2.1.10] describes the details.

- The process involves various minors of the matrix $B(n,r)$ [JN 2.1.10(3.1)]. Lemma 3.1 lists several relations based on Karlin [3, p 59]. Each relation contains 3 minors of order r, and 3 of order $(r-1)$.
- For a particular choice of input determinants we display a tree. Each node in the tree diagram [JN 2.1.10 (3.19)] represents a relation of the form [JN 2.1.10 (3.3)], where the listed minor is the first to appear in that formula. Using these relations, it may be seen that the single minor in the extreme right-hand column of the diagram (i.e. one of the D(n,r) determinants) may be expressed in terms of those minors appearing in the left-hand column (i.e. a set of $\Delta(n,r)$ determinants).
- For the RH case and four of the CG cases (k=1,3,8,24), we have calculated the values of the minors appearing in various trees. In the attached sample files we have listed information about the 20 nodes closest to the right. For each node we give the signs of the 6 minors associated with that node. The report [JN 2.1.10 Sec. 3.5] explains the details of the display.
- The file k81 relates to the $k = 8$ CG case, with $n = 2$, $r = 8 - 76$. The header for each block of 20 lines is n r. Columns 3, 4 contain (j,k) of $M(r,j,k)$. The last 6 columns contain the sign of the corresponding minor, 1 for + and -1 for -. The first of the 6 columns in row 20 is the sign of $D(n,r)$.
- For all values of r shown, the last 3 rows are all positive. For the $k = 8$ CG case file k82 (n=8, r= 68 - 76) the last 11 rows are all positive. For the $k = 8$ CG case file k83 (n=10,12, r=68 -76) the last 15 rows are all positive when $n = 10$; the last 19 rows are all positive when $n = 12$.
- For the $k = 24$ CG case k241 the number of positive rows is 3 (n=2); 5 (n=4); 8 (n=4); 11 (n=8); 15 (n=10).
- For all cases we have studied, namely RH and CG k=1, k=3, k=8, and k=24, the behavior is similar. For a given case and value of n the number of positive rows before end is $\geq$ a constant (i.e. some of the low r cases have more than the standard number of positive rows). For these five cases the data is consistent with the following numbers of positive rows.

RH

n	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
$l(n)$	15	29	41	63	80	99	109	131	168	195	209	239	271	305	379

CG1

n	2	4	6	8
$l(n)$	8	15	24	35

CG3

n	2	4	6	8	10	12
$l(n)$	5	7	8	11	15	19

CG8

n	2	4	6	8	10	12
$l(n)$	3	5	8	11	15	19

CG24

n	2	4	6	8	10
$l(n)$	3	5	8	11	15

- The relevance of these patterns, if any, remains to be discovered. However, it should be noted that not all the information in the Wronskian $w(r, m; t)$ has been used, because of the restriction $t = 0$

References

1. J. B. Conrey and A. Ghosh, *Turán inequalities and zeros of Dirichlet series associated with certain cusp forms*, Trans. Amer. Math. Soc. **342** (1994), 407–419.
2. G. Csordas, T. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541.
3. S. Karlin, *Total Positivity*, Stanford University Press, Stanford, CA, 1968.

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